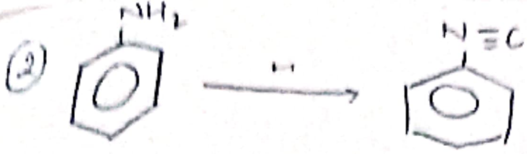
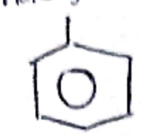
(2) 

Carbon tetrachloride:

$$CS_2 + 3Cl_2 \longrightarrow CCl_4 + S_2Cl_2$$

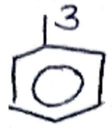
$$CCl_4 + H_2O \longrightarrow COCl_2$$

ARYL HALIDES (Ar+X)



→ Aryl halide

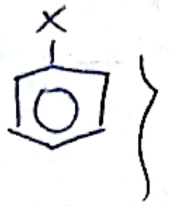
(i) $NaNO_2$: Diazotisation.



→ Not Aryl halide

(ii) $NaNH_2 \xrightarrow[R-X]{}$ $\overset{+}{R}$ gona Attack on Negative Part...

$H_2C=CH-X$ (Vicinal halides)

* 

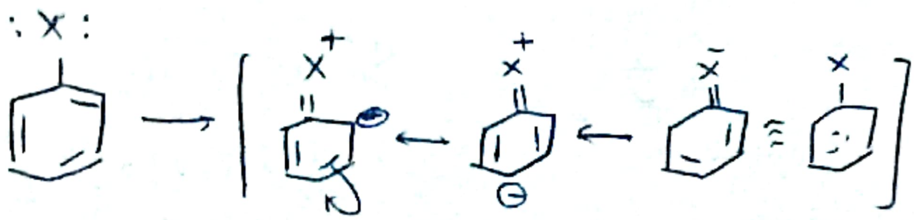
$\xrightarrow{\bar{OH}}$, $\xrightarrow{\bar{OR}}$, $\xrightarrow{NH_3}$, $\xrightarrow{\bar{CN}}$, $\xrightarrow[AlCl_3]{Ar-H}$ } NO Rxn

$R-X + \bar{OH} \rightarrow ROH$

$R-X + CN \longrightarrow R-CN$

⇨ Undergoes NSR

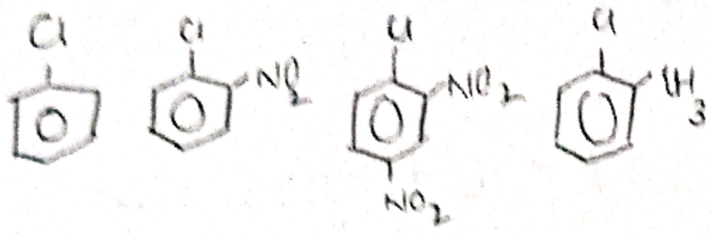
1. Double bond character b/w Carbon & Halogen



2 a bond Strength alkyl halid R-X
Sp3

Aryl :: Sp^2-X BL↓ BS↑

3 Resonance.

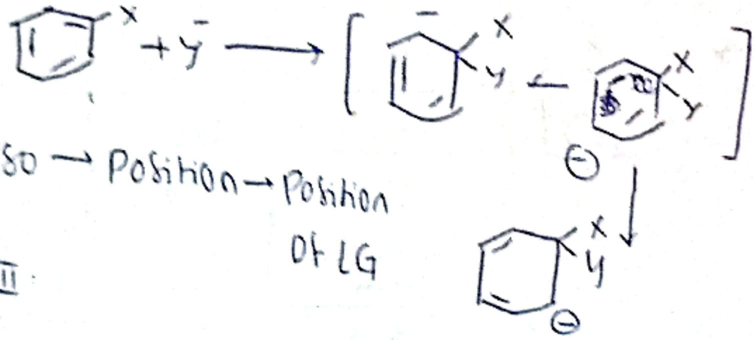


III > II > I > IV

S_NAr : Aromatic Nucleophilic Substitution.

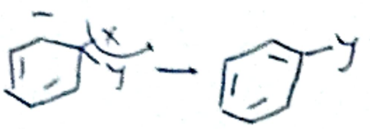
S_N^2 Ar : (Addition, Elimination Reactions)

Two Step :



IPSO → Position → Position
Of LG

Step II :



IPSO Product

⇨ EWG at
Ortho and Para Tse
SNAr Reaction

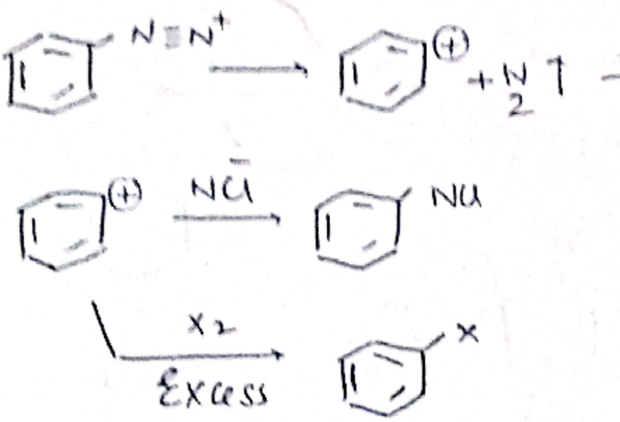
S_N^1 & S_N^2
$\bar{I} > \bar{Br} > \bar{Cl} > \bar{F}$ are all LG

S_N^1Ar & S_N^2Ar
$\bar{F} > \bar{Cl} > \bar{Br} > \bar{I}$ LG

* * * * *

S_N1 Ar Reaction

(Benzene diazonium)



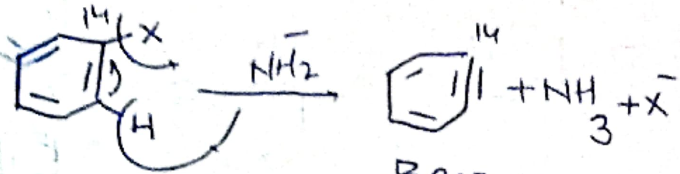
→ Slowest formation $\overset{+}{C}$

Rate = K[Substrate]

Benzyne Mech : (elimination - Addition Mech)

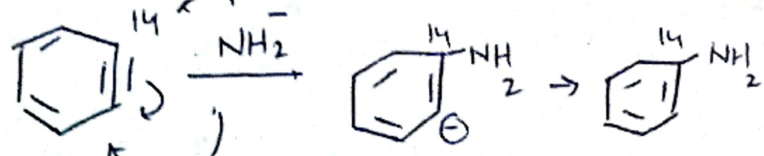
* When there is no effecting group $NaNH_2$/Liq NH_3
* in Presence of Strong base.

Elimination

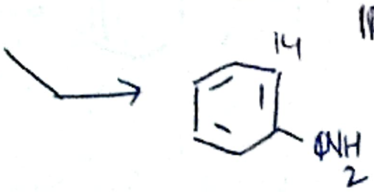


Benzyne

Addition :

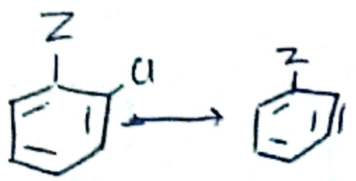


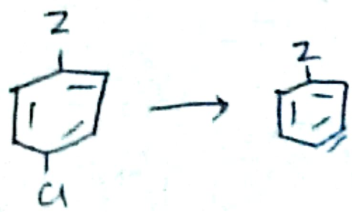
IPSO (It-Self)

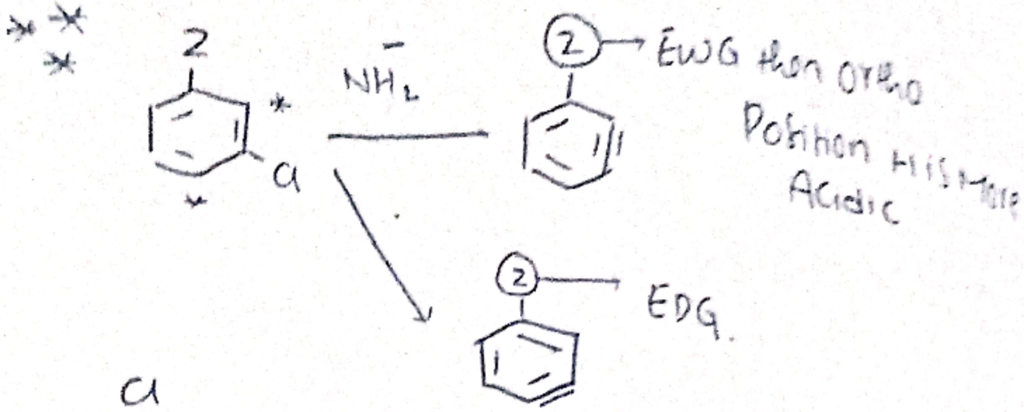


Cine Product

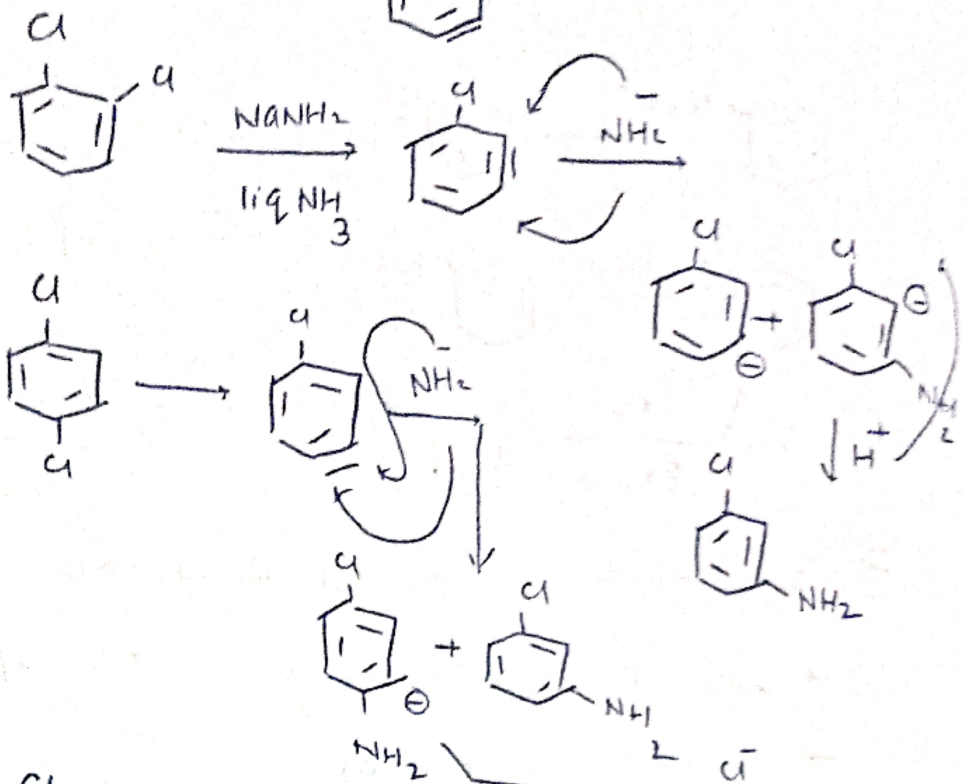
Position which is already occupied by a non hydrogen Substituent

1) 

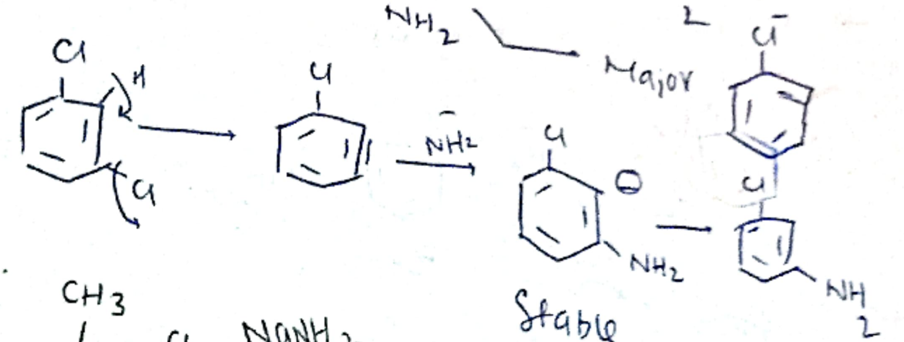
2) 



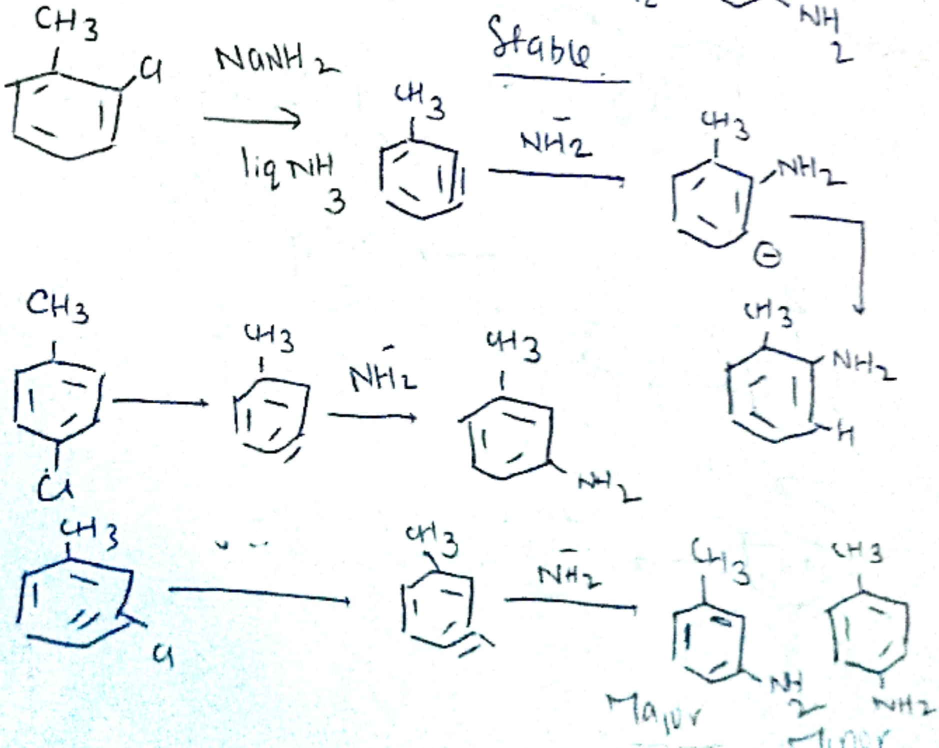
Z
$\bar{NH_2}$
Cl
Z → EWG then ortho Position H is More Acidic
Z → EDG



NaNH2
liq NH3
Major



1.



NaNH2
liq NH3
Stable
Major

	EWG	EDG
OLG	Meta	Ortho
PLG	Para	Meta
MLG	Meta	Meta.